



Ahlam Mallak

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PROFESSIONAL SUMMARY

An enthusiastic **Scientist-Engineer**, with **10+ years** of experience in various topics. I have worked with many reputable **industrial and academic institutions** in the **Middle East, South Korea, and Germany**. I am also a **patent holder** and the author of several other publications. As well as fluent in different natural and programming languages. Seeking to join your respectful institution to leverage my skills and passion and help your institution meet your productivity, creativity and innovation goals.

EXPERIENCE

08/2021 – 08/2023

Research and Development, Goethe University/ Goethe University Hospital, Germany

Responsibilities:

- **Research and development:**
 - Working with applying statistical approaches such as S-LDSC (Stratified LD-Score Regression) to estimate the heritability of numerous genes and cell-types in respect of various cardiovascular diseases.
 - Creating advanced machine and deep learning approaches to estimate the heritability of various functional categories (i.e., genes, enhancers, cell-types...etc.) which overcome the drawbacks of state-of-the-art statistical methods.
- **Data management role:**

I was responsible of the installation, monitoring and maintenance of an online lab notebook to create and exchange experiments between lab members sufficiently.

10/2016 – 03/2021

Research Assistant and Lecturer, Universität Siegen, Germany

Responsibilities:

- **Research and development:** working as a research assistant in a 3-years DFG funded project. The research topic requires the expertise in machine and deep learning, big data analysis, knowledge-management and time-series data analysis to achieve predictive maintenance to distributed systems.
- **Lecturing:** Giving the practical part of the software engineering class using Python to master's students.
- **Master Thesis Supervision:** I supervised three Master theses in the

topics of machine learning, wireless sensor networks, IoT and Bioinformatics.

09/2014 – 08/2016

Research and Development, Chosun University 조선대학교, South Korea

Responsibilities:

- working as a full-time research assistant in the computer vision, multimedia and bioinformatics department, on a Korean National Research (NRF) funded project. The aim of this job is to use machine learning and medical image processing techniques for Alzheimer's disease computer-aided diagnosis.

09/2015 – 06/2016

UNESCO Ambassador of Lebanon, UNESCO, Gwangju-South Korea

Responsibilities:

- **Lecturing and event planning:** conduct many cultural events and presentations in schools of various age groups, to raise the knowledge, awareness and acceptance of different cultures and languages.

07/2013 – 08/2014

E-Learning Courses Developer and Animator, Knowledge Horizon, Jordan

Responsibilities:

- **Programming/Developing:** Program the animated part using action script 2/3, and the web part using Java.
- **Animation:** Creating animations for the online courses using Adobe Flash professional.

2016 - 2021

PhD Universität Siegen, Electrical Engineering and Computer Science

GPA of **1.0**, *Magna Cum Laude*.

Dissertation: **"Comprehensive Machine and Deep Learning Fault Detection and Classification Approaches of Industry 4.0 Mechanical Machineries: With Application to A Hydraulic Test Rig"**

EDUCATION

2014 - 2016

MS Chosun University 조선대학교- South Korea

Computer Engineering and Bioinformatics

GPA of **4.31/4.5** which is equivalent to Excellent.

Thesis: **"Synergistic Structure of SOMs and GMKL Based on MR Images for Alzheimer's Disease Diagnosis."**

2008 - 2013

BS Al Balqa' Applied University

Computer and Electrical Engineering

The Faculty of Engineering Technology

GPA of **3.85/4.0** which is equivalent to Excellent.

Best poster award, MATLAB EXPO 2019 in Munich:

Title: "A Dynamic Sensor Fault Detection and Identification System in IoT using MATLAB and Simulink."

HONORS AND AWARDS

A DFG funded research at Universität Siegen: "Active Diagnosis based on Semantic Web Technologies for Distributed Embedded Real-Time Systems (ADISTES)"

A Korean National Research (NRF) funded project at Chosun University

A Certificate of Excellence awarded from KAIST University in Daejeon-South Korea.

LANGUAGE SKILLS

Arabic: Native (C2)

English: Native/Fluent (C2)

German: Intermediate (B2 Spoken and Written)(Telc B1 Deutsch Certificate)

Korean: Basic (A2) (I have TOPIK certificate level one in Korean)

TECHNICAL SKILLS

Programming & Data Science

Python (Pandas, NumPy, Scikit-learn), R, MATLAB, C++

Machine Learning & AI

Machine Learning, Deep Learning (TensorFlow, PyTorch, Keras), Predictive Modeling, Time-Series Analysis

Data Engineering & Management

SQL, MySQL, Data Integration, Data Cleaning & Transformation, Reproducible Workflows

Cloud & Visualization

Microsoft Azure, Power BI, Streamlit

Tools & Systems

Linux/Unix, Git, Data Pipelines, Electronic Lab Notebooks

CORE COMPETENCIES

Bioinformatics • Data Analysis • Machine Learning • Data Integration • Problem Solving • Team Collaboration • Communication • Research & Innovation • Technical Writing

PUBLICATIONS

Patent:

- Mallak and S.-W. Lee, "Method for Representing Graph-based Block-minutiae for Fingerprint and the Fingerprint Recognition System by using the Same", Application No. 10-2016-0008988, January 2016.

Theses:

- Master thesis at Chosun University in South Korea: "Synergistic Structure of SOMs and GMKL Based on MR Images for Alzheimer's disease Diagnosis (알츠하이머병 진단을 위한 MRI 영상기반 SOM 과 GMKL의 상승 결합 구조)"
- PhD thesis at Universität Siegen: "Comprehensive Machine and Deep Learning Fault Detection and Classification Approaches of Industry 4.0 Mechanical Machineries: With Application to A Hydraulic Test Rig."

Journals:

- Mallak and S.-W. Lee, "The Effect of PCA On Kernel Classifiers for the Diagnosis of Alzheimer's Disease," Journal of the Korea Institute of Next Generation Computing, Vol. 12, No. 4, April 2016, pp. 25-32 (in English).
- Mallak and M. Fathi, "A Hybrid Approach: Dynamic Diagnostic Rules for Sensor Systems in Industry 4.0 Generated by Online Hyperparameter Tuned Random Forest," Sci, vol. 2, no. 4, Art. no. 4, Dec. 2020, doi: 10.3390/sci2040075.
- A. Mallak and M. Fathi, "Sensor and Component Fault Detection and Diagnosis for Hydraulic Machinery Integrating LSTM Autoencoder Detector and Diagnostic Classifiers," Sensors, vol. 21, no. 2, Art. no. 2, Jan. 2021, doi: 10.3390/s21020433.
- (Publishing in progress)
Mallak, A.; Fathi, M. Unsupervised Feature Selection Using Recursive k-Means Silhouette Elimination (RkSE): A Two-Scenario Case Study for Fault Classification of High-Dimensional Sensor Data. Preprints 2020, 2020080254 (doi: 10.20944/preprints202008.0254.v1).

Conferences:

- Mallak, M. Schulz, "Combining Epigenome and Genetic Mutation Data from GWAS for the Study of Disease-Relevant Cell Types", ECCB2022 21st European conference on computational biology, Sitges Spain, 2022.
- A. Mallak, M. Fathi, "SenGen: A Two-Phase Dynamic Simulation and Toolbox of An Indoor Mobile Wireless Sensor Network for Sensor Monitoring and Dataset Generation", 2019 IEEE Computational Science & Computational Intelligence, Las Vegas, 2019.
- A. Mallak, C. Weber, M. Fathi and A. Holland, "Active diagnosis automotive ontology for distributed embedded systems," 2017 IEEE European Technology and Engineering Management Summit (E-TEMS), Munich, 2017, pp. 1-6. doi: 10.1109/E-TEMS.2017.8244219
- A. Mallak, A. Behravan, C. Weber, M. Fathi and R. Obermaisser, "A Graph-Based Sensor Fault Detection and Diagnosis for Demand-Controlled Ventilation Systems Extracted from a Semantic Ontology," 2018 IEEE 22nd International Conference on Intelligent Engineering Systems (INES), Las Palmas de Gran Canaria, 2018, pp. 000377-000382. doi: 10.1109/INES.2018.8523895
- A. Behravan, A. Mallak, R. Obermaisser, D. H. Basavegowda, C.

Weber and M. Fathi, "Fault injection framework for fault diagnosis based on machine learning in heating and demand-controlled ventilation systems," 2017 IEEE 4th International Conference on Knowledge-Based Engineering and Innovation (KBEI), Tehran, 2017, pp. 0273-0279. doi: 10.1109/KBEI.2017.8324986

- Mallak, J. Gwak, and S.-W. Lee, "Computational and Comparative Study on Multiple Kernel Learning Approaches for the Classification Problem of Alzheimer's disease," Proceedings of The International Conference on Nature of Computation and Communication, Rach Gia, Vietnam, March 2016.
 - Mallak and S.-W. Lee, "Effect of PCA on SVM classifier and k-means Clustering for Alzheimer's Disease Diagnosis," Proc. of 2015 KING Fall Conference, Daejeon, Korea, November 2015, pp. 68-71.
 - I. K. Virdaus, A. Mallak, S.-W. Lee, G. Ha, and M. Kang, "Fingerprint Verification with Crossing Number Extraction and Orientation-Based Matching," Proceedings of The International Conference on Next Generation Computing, Bangkok, Thailand, February 2016, pp. 113-115.
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